THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 4: **Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x + y = 12$ and $x^2 + y^2 = 90$, then $xy =$ 1

(a) 28 **(b)** 27 **(c)** 26 **(d)** 29

2. If $x - y = 4$ and $x^2 + y^2 = 40$, then $xy =$ 1

(a) 11 **(b)** 12 **(c)** 14 **(d)** 13

3. If $x^2 + \frac{1}{x^2} = 34$, then $\left(x + \frac{1}{x}\right)^2 =$ 1

(a) 35 **(b)** 38 **(c)** 36 **(d)** 37

4. $x^3 - 64$ factorises as: 1

(a) $(x + 4)(x^2 - 4x + 16)$ **(b)** $(x - 4)(x^2 + 4x + 16)$ **(c)** $(x - 4)(x^2 - 4x + 16)$ **(d)** $(x - 4)^3$

5. If $x + y = 9$ and $xy = 20$, then $x^3 + y^3 =$ 1

(a) 191 **(b)** 190 **(c)** 189 **(d)** 188

6. If $x - y = 4$ and $xy = 6$, then $x^3 - y^3 =$ 1

(a) 135 **(b)** 137 **(c)** 136 **(d)** 138

7. If $x + \frac{1}{x} = 6$, then $x^3 + \frac{1}{x^3} =$ 1

(a) 198 **(b)** 197 **(c)** 199 **(d)** 200

8. The value of 98^3 is: 1

(a) 941192 **(b)** 941191 **(c)** 941193 **(d)** 941194

9. **Assertion (A):** If $x + y = 7$ and $xy = 10$, then $x^2 + y^2 = 29$. 1

Reason (R): $x^2 + y^2 = (x + y)^2 - 2xy$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** $(2x-3)^3 = 8x^3 - 36x^2 + 54x - 27$. 1

Reason (R): $(a+b)(a-b) = a^2 - b^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Factorise $x^3 + 125$. 2

12. If $x + y = 11$ and $xy = 24$, find $x^3 + y^3$. 2

13. Find the value of 95^3 using an identity. 2

14. Factorise $64x^3 + 27$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If $x + \frac{1}{x} = 11$, find $x^3 + \frac{1}{x^3}$. 3

16. Factorise $x^2 - 7x + 6$ and use it to find the value when $x = 8$. 3

17. Factorise $64x^3 - 1$. 3

OR

If $x + \frac{1}{x} = 12$, find $x^3 + \frac{1}{x^3}$. 3

18. Factorise $x^2 - 8x + 16$ and use it to find the value when $x = 9$. 3

OR

Factorise $8x^3 - 125$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A toy maker in Channapatna makes wooden cubes. One cube has edge $(2x + 1)$ cm and another has edge $(3x - 2)$ cm. 5

(i) Expand the volume of the first cube.

(ii) Find the volume of the first cube when $x = 2$.

(iii) Use $(10 + 1)^3$ to find 11^3.

(iv) Expand the volume of the second cube and verify the result at $x = 1$.

20. If $x - y = 1$ and $xy = 42$, find $x^2 + y^2$, $x^3 - y^3$ and $(x + y)^2$. 5

OR

If $x + y = 11$ and $x^2 + y^2 = 61$, find xy, $(x - y)^2$ and $x^2 - y^2$ given that $x > y$. 5

END OF QUESTION PAPER — **Total: 40 Marks**